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Grand Gulf Nuclear Station
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10 CFR 50.73

GNRO2025-00014

April 16, 2025

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Grand Gulf Nuclear Station, Unit 1 Licensee Event Report 2025-001-00,
Manual Scram Due to Degraded Condenser Vacuum

Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
Renewed License No. NPF-29

Attached is Licensee Event Report (LER) 2025-001-00, Manual Scram Due to Degraded Condenser Vacuum. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(iv)(A), for events or conditions that resulted in the manual or automatic actuation of the reactor protection system.

This letter contains no new Regulatory Commitments. Should you have any questions concerning the content of this letter, please contact me at 802-380-5124.

Sincerely,

Jeff Hardy
Digitally signed by Jeff Hardy
DN: cn=Jeff Hardy, o=Grand Gulf
Nuclear Station, ou=Manager,
Regulatory Assurance,
email=jhardy@entergy.com
Date: 2025.04.16 09:52:49 -05'00'

JAH/saw

Attachments: Licensee Event Report 2025-001-00

cc: NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

U.S Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Attachment
Licensee Event Report 2025-001-00



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Grand Gulf Nuclear Station, Unit 1

☒ 050☐ 052

2. Docket Number

00416

3. Page

1 of 2

4. Title

Manual Scram Due to Degraded Condenser Vacuum

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
02	15	2025	2025	001	00	04	16	2025	N/A	☐ 050
									N/A	☐ 052

9. Operating Mode

1

10. Power Level

84

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)(i)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ Other (Specify here, in abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Jeff Hardy, Regulatory Assurance Manager

Phone Number (Include Area Code)

802-380-5124

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
				N					

14. Supplemental Report Expected

☒ No☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month

Day

Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On February 15, 2025, at 2240 CT, Grand Gulf Nuclear Station, Unit 1 operators initiated a manual Reactor Protection System (RPS) actuation due to degrading condenser vacuum which resulted from a loss of seal steam. Rainwater intrusion had entered the enclosure building due to degraded roofing. The water entered a load control center which caused an overcurrent and loss of power in a static inverter. The loss of the static inverter resulted in a loss of seal steam that led to a degraded condenser vacuum. All control rods were fully inserted with no complications.

The cause of the event was determined to be water intrusion due to degraded roofing of the enclosure building.

This report is being made in accordance with 10 CFR 50.73(a)(2)(iv)(A) for any event that resulted in manual actuation of the RPS system.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Grand Gulf Nuclear Station, Unit 1	☒ 050	00416	YEAR	SEQUENTIAL NUMBER	REV NO.
	☐ 052		2025	001	00

NARRATIVE**Plant Conditions:**

At the time of the event Grand Gulf Nuclear Station (GGNS), Unit 1 was in MODE 1 at 84 percent power. The Division 2 Emergency Diesel Generator was inoperable due to scheduled maintenance, and all other safety systems were operable.

Description of Event:

On February 15, 2025, rainwater intrusion entered the enclosure building due to degraded roofing. The water entered a load control center which caused an overcurrent and loss of power in a static inverter.

At 2237 CT, the loss of power to the static inverter caused a loss of seal steam and degrading condenser vacuum. At 2240 CT, a manual Reactor Protection System (RPS) actuation was initiated due to degrading condenser vacuum. All control rods fully inserted, and there were no complications.

Reportability:

This report is being made in accordance with 10 CFR 50.73(a)(2)(iv)(A) for any event that resulted in manual actuation of the RPS system.

Cause(s):

The direct cause of the event was water intrusion due to degraded enclosure building roofing which entered the load control center and caused an overcurrent condition and loss of power to the static inverter.

Corrective Actions:**Completed Actions:**

- Restored the function of the enclosure building roof drains.
- Replaced damaged Load Center Bus components

Planned Actions:

- Replace enclosure building roof.
- Update preventative maintenance instructions for roof inspections to ensure that degradation is identified.
- Review all single power source inverters to identify and correct any single point vulnerabilities.

Safety Significance:

There were no actual safety consequences associated with this condition. Plant systems responded as designed and the unit was brought to Mode 3 by Control Room Operators.

Previous Similar Events:

None identified.